MILITARY SPECIFICATION

SCRAPER-TRACTORS: EARTH MOVING, HYDRAULIC OPERATED, 14 CUBIC YARD MINIMUM STRUCK CAPACITY, DED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-propelled, diesel-engine-driven, hydraulic operated, 14 cubic yard (cu yd) minimum struck capacity, earth moving, scraper-tractors.

1.2 Clasification. Scraper-tractor shall be one of the following types as specified (see 6.1):

Type I - Two Driven Traction Wheels, 4 by 2.
Type II - Four Driven Traction Wheels, 4 by 4 or 6 by 4.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage, Vehicular, Ignition, Lighting, and Starting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3805

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MIL-P-514 - Plate, Identification, Instruction and Marking, Blank.
MIL-T-3351 - Tractor, Full-Tracked, Low-Speed; Tractor Wheeled, Agricultural; and Tractor Wheeled, Industrial; and Their Attachments, Packaging of.

STANDARD

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E380 - Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard commercial product. The scraper-tractor shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the scraper-tractor being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.1), the contractor shall furnish a scraper-tractor for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

* 3.4.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as

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specified (see 6.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ASTM E380.

3.5 Ease of maintenance. Design of the scraper-tractor shall permit ready accessibility to all parts for maintenance and repair in the field. Replacement and adjustment of component parts and accessories shall be accomplished with minimum drainage requirements and disturbance to other elements of the scraper-tractor.

3.6 Safety. All moving parts, parts subject to high operating temperatures, and parts energized electrically which are of such a nature or so located as to become a hazard to operating and maintenance personnel shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating functions. Platforms, steps, handholds, and rails shall be furnished and located, where necessary, to assure safe, easy access by personnel to those areas requiring frequent maintenance or service. All safety systems provided for the operator shall be in accordance with DoL, OSHA standards. Scraper-tractor shall be provided with at least the following safety equipment:

a. Windshield with power operated windshield wiper(s).
b. Horn.
c. Backup alarm device conforming to SAE J994.
d. Seat belt for operator's seat.
e. Rollover protective structure at operator's station.
f. Fenders over all driving wheels, front and rear.

* 3.7 Power to weight ratio. The ratio of the weight of the scraper-tractor with 48,000 pound (lb) payload to the net flywheel horsepower (hp) of the engines(s) shall be not more than 345 lb per hp, for the 4 by 2 type, and not more than 280 lb per hp, for the 4 by 4 type, and not more than 400 lb per hp, for the 6 by 4 type.

* 3.8 Dimensions, weights, and capacities. Scraper-tractor shall conform to the dimensions, weights, and capacities specified in table I (see 3.4.1). The struck and heaped capacity yardage ratings shall be in accordance with SAE J741.

TABLE I. Dimensions, weights, and capacities.

Struck capacity, cu yd (not less than)	14
Heaped capacity, cu yd (not less than)	20
Payload, lb (not less than)	48,000
Cutting width, inches (not less than)	108
Cutting depth, inches (not less than)	11
Spreading depth, inches (not less than)	17
Nonstop turning width, feet (not more than)	48

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3.9 Performance. Scraper-tractor shall withstand the hard usage encountered, without permanent deformation or damage, while performing the following operations:

a. The type I scraper-tractor shall load undisturbed or recompacted soil (see 6.4.1), to the struck capacity with assistance from auxiliary equipment.
b. The type II scraper-tractor shall load undisturbed or recompacted soil to the struck capacity in not more than 1 minute without assistance from auxiliary equipment.
c. With a minimum heaped payload as specified in table I, travel at an average speed of not less than 23 miles per hour (mph) on maintained earth haul roads (see 6.4.2).
d. With a minimum heaped payload as specified in table I, negotiate a grade having a minimum slope of not less than 20 percent.
e. Without payload, traverse a longitudinal grade having a minimum slope of not less than 30 percent.
f. Dump and spread the soil without assistance from auxiliary equipment.

3.9.1 Environmental conditions. The scraper-tractor engine(s) shall be capable of starting and satisfactorily performing at a temperature from -20 degrees Fahrenheit (°F) to +125°F. When an ether priming system is required, it shall be of the measured shot type with storage capacity of not less than 12 fluid ounces.

* 3.10 Tractor frame. The tractor frame shall support or incorporate as integral parts the towing device(s), crankcase guard, fenders, oscillating hitch assembly, operator's stations, powerplant, and power train. Frame shall be mounted on two or four power driven pneumatic-tired wheels. Towing device(s) shall withstand a pulling force equivalent to not less than 150 percent of the scraper-tractors gross vehicle weight (gvw) (see 6.4.3). Crankcase guard shall protect the crankcase, flywheel housing, accessories, and other assemblies exposed to damage from projecting obstuctions likely to be encountered during normal earth moving operations.

* 3.11 Draft frame. The draft frame consists of the hitch assembly housing, gooseneck, and yoke. Front of the frame shall be coupled to and supported by the tractor; rear shall be mounted on scraper bowl with pneumatic-tired wheels. The draft frame shall be equipped with a mechanical locking device to prevent tractor/scraper articulation during lifting operations and maintenance functions.

3.12 Push frame. The push frame shall incorporate a push plate or roller, and towing hook(s) and shall be mounted on the rear of the scraper. The frame shall support the rear powerplant and power train (if applicable). The push plate or roller and frame shall withstand the forces applied to it without deformation or breakage when the scraper-tractor is push-loaded with another scraper-tractor or tracked tractor of equivalent size.

* 3.13 Scraper bowl. The scraper bowl shall be of the front-loading, front-lifted, open-bowl type, complete with apron, blade base, ejector, cutting edges, and side bits. The apron, ejector, and bowl shall be hydraulically operated. The bowl shall be power raised and power lowered and

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capable of holding any position through its travel range with any load throughout the capacity range. The bowl shall be free of overhead beams, structures, and hydraulic components that would obstruct loading by means of crane shovels, draglines, and loaders. The bowl shall withstand, without permanent deformation or damage, all operating forces of the apron and ejector. When tested as specified in 4.3.1.2, the bowl shall carry a load of * 48,000 lb without permanent deformation or damage and without deflecting in overall width to an extent which would adversely affect operation or cause excessive rub plate wear. In no case shall the overall width deflection exceed 1/2 inch.

3.13.1 Cutting edges. Cutting edges shall consist of not less than three sections or more than four sections. All sections shall be reversible with two sections interchangeable. Cutting width shall be as specified in table I.

3.13.2 Side bits. Side bits shall be reversible side-to-side and interchangeable.

* 3.14 Axles. The type I scraper-tractor shall be a 4 by 2 combination with a single engine on the tractor with drive axle coupled to tow the scraper on nondriven wheels. The type II scraper-tractor shall be either a 6 by 4 combination type with a single engine on a tractor with 4-wheel drive coupled to tow a scraper with nondriven wheels, or by 4 by 4 combination type driven by an engine on the tractor with a drive axle and another engine on the back end of scraper with a drive axle for pushing. The drive axle of the tractor shall be equipped with either a manually controlled differential lock, no spin differential, or automatic torque proportioning differential. If the scraper has a driven axle, it shall be equipped with an automatic differential lock. Planetary positive reduction gearing or direct spur gear reduction gearing shall be utilized for all drive axles.

3.15 Transmission(s) and torque converter(s). The scraper-tractor shall have full torque power shift transmission(s) with lockup in at least the top four speeds, filter(s), and torque converter(s). The transmission system shall provide not less than five forward and one reverse speed. For the 4 by 4 scraper-tractor with two engines, both the tractor and scraper transmissions shall be controlled and shifted together by a single lever. Torque converter oil coolers shall be furnished. Coolers shall maintain the oil entering the converters at a temperature not exceeding 275°F. Tractor torque converter oil temperature indicators, or high temperature warning devices located at the operator's station, shall be furnished, and the scraper-tractor torque converter oil temperature may be monitored by audio warning system.

3.16 Steering mechanism. Steering mechanism for the scraper-tractor shall be power steering or power assisted steering. The steering mechanism shall be capable of turning the tractor 90° right or left of the longitudinal centerline of the scraper.

3.17 Brakes.

3.17.1 Service brakes. The scraper-tractor shall be equipped with full air or combination air-over-hydraulic service brakes on all wheels, stopping the fully loaded scraper-tractor from a speed of 20 mph within 90 feet from

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the point of brake pedal application. Brakes shall be equipped with an emergency stopping system to provide braking in the event of air pressure loss. Air reservoir(s) shall have moisture bleeder(s).

3.17.2 Parking brake. The scraper-tractor shall be equipped with a parking brake capable of holding the fully loaded scraper-tractor on a 15 percent grade without the assistance of the service brakes.

3.18 Wheels, rims, and tires. The tires shall be of same size and interchangeable front to rear. Tires shall be wide base, tubeless, with rock type tread. Tire and rim sizes, ply rating, and load rating at 30 mph shall conform to TRA recommendations. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with scraper-tractor loaded to rated gvw (with overload not to exceed 6 percent) (see 6.4.3).

3.19 Engine(s). The scraper-tractor shall be powered by industrial type diesel engine(s). The power and speed of the engine(s) shall be sufficient to meet the performance requirements specified herein. When two engines are furnished, the engines shall be the same make. The engine(s) shall be furnished complete with the following accessories:

a. Two stage intake air cleaner with service indicator.
b. Full flow oil filter with bypass.
c. Fuel filter or filters.
d. Hour meter (front engine only).
e. Fuel tank with sufficient capacity for 8 hours operation.
f. Lubricating oil pressure indicator.
g. Coolant temperature indicator.
h. Electric cranking system (hydraulic or air starting acceptable for rear engine).
i. Battery(s) (see 3.19.1).
j. Battery charging alternator/rectifier system.
k. Battery charging indicator.
l. Torque converter oil temperature indicator.
m. Radiator guard.
n. A tachometer (front engine only).

* 3.19.1 Dry charged batteries. When specified (see 6.1), batteries shall be dry charged in accordance with W-B-131, without electrolyte, with sealed caps to prevent the intrusion of atmospheric moisture.

3.19.2 Exhaust. The engine(s) shall be equipped with exhaust muffler and stack. Exhaust stack outlet shall be provided with rain caps, 45° or 90° bends, or moisture drain.

3.20 Lighting. The lighting system shall be either 12 or 24 volt, but shall be compatible with the voltage of battery(s) and alternators including the following:

a. Two headlamps and two forward directed floodlamps or four headlamp bright and dim system.
b. One scraper floodlamp directed to illuminate the bowl, apron, and cutting edge.
c. Instrument panel lamp(s) to illuminate all instruments.

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* 3.21 Hydraulic system. A hydraulic system shall be utilized for transmitting engine power to the mechanisms that operate the apron, bowl, ejector, and steering. The hydraulic system shall be complete with all operating accessories, including pump(s), reservoir(s), valves, filters, hoses, piping, fittings, and pressure relief valve(s). Pump(s) shall supply a continuous flow of hydraulic fluid, and shall have sufficient capacity to energize all hydraulically powered components. The hydraulic system shall be capable of withstanding a bursting pressure of four times the working pressure. The system, including hoses, shall be suitable for operating temperatures ranging from -20°F to 250°F. Hoses shall be protected against chafing through utilization of supports and protective devices. Hydraulic system shall be protected against the entrance of dirt or other foreign material. A wiper shall be provided on each hydraulic cylinder located near the piston rod seal to prevent dirt and abrasives from damaging the seal or entering the cylinder. Hydraulic system shall be furnished with full flow filter(s) having a micron rating not greater than 25. When an open type hydraulic system is furnished, reservoir(s) shall be vented by a separate vent or through the filler cap. In either case, the vent shall be equipped with a micronic filter element to prevent dirt from entering the reservoir and shall have a micronic rating of not greater that 25. Reservoir(s) shall have sufficient capacity to allow for maximum displacement of all cylinders at the same time, and to aid heat dissipation. Return line shall be terminated below the minimum oil level of the reservoir. Suction line shall be above the reservoir bottom. On open type hydraulic systems, the suction line shall be protected with a readily accessible strainer. A valve with three separate operating sections and individual control levers shall be provided to permit independent control of the apron, ejector, and bowl in both directions. The apron valve shall provide control for power raise, lower, hold, and float. The bowl valve shall permit control for power raise, power lower, and hold. The ejector valve shall allow control for power eject, hold, return, and when ejector return is hydraulically operated, an automatic disconnect and return.

3.22 Operator's station. Operator's station shall consist essentially of a shock absorbing operator seat, steering wheel, a safety glass windshield equipped with power operated windshield wiper(s), instrument panel, operating controls, and a rollover protective structure conforming to SAE J1040. The seat and backrest shall be covered with a durable waterproof covering. The seat shall be equipped with a seat belt conforming to SAE J386. All controls and instruments that are not already identified to show their intended funtion or operation shall be clearly identified for their intended operation and funtion with permanently affixed identification plates of noncorrosive material. Decals shall not be used. Controls shall be located within the dimensional spaces specified in SAE J898. Instruments shall be readily visible to the operator when seated in a normal operating position. Instruments shall be protected against damage by an anti-vandalism panel guard, unless the scraper-tractor is equipped with an enclosed cab with lockable door.

* 3.22.1 Cab. When specified (see 6.1), a fully enclosed cab of all steel or fiberglass construction shall be furnished. The cab shall have a hinged door and shall enclose the operator's seat and all controls, and shall not obstruct the view of the operator while seated. Closed cab accessories shall include a rearview mirror of the adjustable truck type, with minimum size of 4 by 8 inches, mounted on the left exterior of the cab. The enclosed cab

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shall be furnished with a heater and defroster. The heater and defroster shall be the manufacturer's standard for maintaining the operator's comfort in temperatures down to -20°F. Electric or air powered windshield wipers for front and rear windshields, with separate controls, shall be furnished.

3.23 Toolbox. When specified (see 6.1), scraper-tractor shall be provided with a toolbox made of sheet steel. The toolbox shall have a hinged lid which shall open not less than 90° and shall be provided with a latching device. Toolbox shall be securely fastened to the scraper-tractor in a protected and accessible location. The toolbox shall be of sufficient size to hold the tools normally used to maintain the scraper-tractor.

3.24 Lifting and tying down attachments. When specified (see 6.1), scraper-tractor shall be equipped with lifting and tying down attachments. Attachments shall conform to MIL-STD-209, type III. Attachments shall be indicated by a transportation plate conforming to MIL-P-514, except tying down attachments may be identified by stenciling or other suitable marking of the unit. Tying down marking shall clearly indicate that the attachments are intended for tying down of the scraper-tractor on the carrier when shipped. Transportation plate shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachments, and the required length and size of each sling cable. A silhouette of the scraper-tractor showing the center of gravity shall be provided on the transportation plate.

3.25 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor shall stamp the necessary data in the blank spaces thereon and securely affix said plate in a conspicuous place on the scraper-tractor. Nonferrous screws, rivets, or bolts of not less the 1/8 inch in diameter shall be used to affix the plates. Nomenclature shall be: "SCRAPER-TRACTOR, HYDRAULIC OPERATED, 14 CU YD MIN STRUCK CAPACITY, DED."

* 3.26 Instruction marking. The scraper-tractor shall be equipped with instruction marking suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Marking shall be of a material which will last and remain legible for the life of the equipment. Marking shall be securely affixed to the equipment as per manufacturer's standard procedure.

3.27 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.28 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specifed herein. The color of the finish coat shall be as specified (see 6.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or

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other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturere's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.29 Workmanship.

3.29.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.29.2 Bolted connections. Bolted holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.29.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.29.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on scraper-tractor when a first article is required (see 3.2 and 6.1). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified, the preproduction pack inspection of 4.5 (see 4.5 and 6.1). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5.

4.3 Examination. Each scraper-tractor shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article scraper-tractor shall be tested as specified in 4.4.1, and each production scraper-tractor shall be subjected to the operational test specified in 4.4.2.

4.4.1 First article tests. The first article scraper-tractor shall be subjected to the tests specified in 4.4.1 through 4.4.1.4. Failure to pass one or more tests shall be cause for rejection.

4.4.1.1 Performance test. The scraper-tractor shall load and dump undistrubed or recompacted soil for not less than 8 hours or more than 10 hours. This tests shall consist of approximately 100 load-haul-dump cycles with a one way haul distance of not less the 1500 feet. Each load shall be not less than the SAE J741 struck capacity rating with not less than one half of the loads being the minimum heaped capacity as specified in table I. The type I scraper-tractor shall be push-loaded to achieve required capacities with the type II scraper-tractor being push loaded to achieve heaped capacities only. All hauling and traveling shall be at maximum safe speed. During this test, scraper-tractor shall be observed to determine conformance to:

a. The cutting depth and width, spreading depth, and nonstop turning width specified in table I.
b. Performance and operational requirements specified in 3.9, 3.15, 3.16, 3.21, and 3.22.

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The following separate tests shall be performed:

a. The type II scraper-tractor shall load undistrubed or recompacted soil (see 6.4.1) to the struck capacity for an average of not more than 1 minute for 5 cycles without assistance from auxiliary equipment.
b. With a minimum payload, as specified in table I, scraper-tractor shall negotiate a well compacted earth grade having a minimum slope of 20 percent. Length of the grade shall be not less than twice the length of the scraper-tractor. Scraper-tractor shall be driven up and down the grade not less than ten times each way.
c. Scraper-tractor with minimum payload shall be stopped not less than five times going up and five times going down the 20 percent grade.
d. Without payload, scraper-tractor shall traverse longitudinally along a grade having a minimum slope of not less than 30 percent. Grade shall be of compacted earth having a minimum length of not less than twice the scraper-tractor's length. The scraper-tractor shall make not less than two passes; one from right to left and one from left to right.

* 4.4.1.2 Scraper bowl test. The bowl width shall be measured at 1-foot intervals both horizontally and vertically. Bowl shall be loaded to a capacity of not less than 48,000 lb. The scraper-tractor shall be driven for not less than 4 hours. Load shall be ejected and the bowl measured again for deflection to verify conformance to 3.13. To ascertain conformance to the requirements of 3.13, the bowl, apron, ejector, and load holding devices shall be subjected to the forces specified therein.

4.4.1.3 Brake tests. The service brakes and parking brake shall be tested to determine conformance to 3.17.1 and 3.17.2.

4.4.1.4 Lifting and tying down attachment test. When furnished, lifting and tying down attachments shall be tested to determine conformance to 3.24. The manufacturer may furnish certified engineering calculations in lieu of a test to prove that the lifting and tying down attachments offered meets the requirements set forth im MIL-STD-209.

4.4.1.5 Production sample. Upon acceptance of the first article, the first article shall remain at the manufacturing facility as a production sample, and shall be reconditioned, including replacement of abnormally worn parts and paint touchup or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

4.4.2 Operational test. Each scraper-tractor shall be completely assembled, properly adjusted, and otherwise serviced for operation. All controls shall be operated as many times as necessary to determine ease of operation, effectiveness, and responsiveness, and that the mechanisms actuated by the controls operate promptly, fully, and without restriction, malfuntion, or excessive vibration. Gearcases, bearings, and other moving parts shall be tactually examined for excessive heating or abnormal operation insofar as is practicable and possible. Entire scraper-tractor shall be examined for evidence of failure, slippage, or deformation. Failure to pass any phase of this test shall be cause for rejection.

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4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-T-3351. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.1), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.1).

* 5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-T-3351, type II with the level of preservation and the level of packing as specified (see 6.1).

6. NOTES

* 6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of scraper-tractor required (see 1.2).
c. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
d. When dry charged batteries required (see 3.19.1).
e. When US or International System of Units Measurement (Metric) is required (see 3.4.1).
f. When enclosed cab is required (see 3.22.1).
g. When toolbox is required (see 3.23).
h. When lifting and tying down attachments are required (see 3.24).
i. Color of finish coat (see 3.28).
j. When a preproduction pack inspection is required and time frame repaired for submission (see 4.2.1 and 4.5).
k. Level of preservation and level of packing required (see 5.1).

* 6.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Definitions.

6.4.1 Recompacted soil. Recompacted soil is soil that has been recompacted to not less than 75 percent Standard Proctor and within ±2 percent optimum moisture.

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6.4.2 Maintained earth haul road. A maintained earth haul road has a smooth, firm, level, earth surface that undulates or slightly flexes under load, and is watered and maintained regularly.

6.5 Definitive specification part number. The specification part number is a definitive part number which corresponds to the types of scraper-tractor covered by this specification and defines the requirements of the options presented under this specification. The specification number, and the type code number are combined to form the definitive specification part number.

6.5.1 Cataloging data. For cataloging purposes, part numbers for the scraper-tractor are assigned as follows:

M28632 X

Specification part number. _____
Type code number. _____

6.5.2 Type code number. The type of the scraper-tractor (see 1.2) is identified by a single digit number as follows:

Type I - 1.
Type II - 2.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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